

EFC- Λ CDM Comparative Test Specification v1.1

Pre-Registered Implementation Protocol for DES Y6 and KiDS Legacy Cosmic Shear

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April 2026 (v1.0), updated 2026-04-21 (v1.1)

DOI: [10.6084/m9.figshare.32080083](https://doi.org/10.6084/m9.figshare.32080083) Status: PUBLISHED v1.1 License: CC-BY-4.0

Purpose and Scope

This document specifies — with sufficient precision for external implementation — a three-stage cosmological test of Energy-Flow Cosmology (EFC) against Λ CDM on Stage-III cosmic shear data. The specification is locked at the content-hash level: any implementer following this document exactly will produce reproducible results whose comparison to the pre-registered expectations constitutes a falsifiable test.

This document is NOT:

- Executable code ready to run
- A tutorial on MGCAMB, MGCLASS, Cobaya, or CosmoSIS
- A complete replacement for expertise in cosmological pipelines

This document IS:

- A model specification (μ , Σ functional forms, priors, locked parameters)
- A protocol specification (scan ranges, nuisance treatment, scale cuts)
- A reporting specification (what must be output regardless of outcome)
- A falsification specification (binding stopping and retraction rules)

Implementation requires MGCAMB or MGCLASS access, tested DES Y6 and KiDS Legacy likelihood modules in Cobaya or CosmoSIS, and CPU/cluster resources for MCMC sampling.

1 Model Specification

1.1 Λ CDM Baseline (Reference Model)

Standard six-parameter Λ CDM. Free cosmological parameters: $\{\omega_b = \Omega_b h^2, \omega_c = \Omega_c h^2, 100 \theta_{\text{MC}}, \tau_{\text{reio}}, \ln(10^{10} A_s), n_s\}$.

Priors (DES Y6 fiducial, identical for both models):

Parameter	Prior type	Range
ω_b	Uniform	[0.020, 0.025]
ω_c	Uniform	[0.09, 0.15]
$100 \theta_{\text{MC}}$	Uniform	[1.03, 1.05]
τ_{reio}	Gaussian	0.054 ± 0.007
$\ln(10^{10} A_s)$	Uniform	[2.0, 4.0]
n_s	Uniform	[0.87, 1.07]

Derived parameter: $S_8 \equiv \sigma_8 \cdot (\Omega_m/0.3)^{0.5}$.

1.2 EFC Model

Two additional parameters $\{A_\mu, A_\Sigma\}$ extending Λ CDM. No other EFC parameters float.

EFC (μ, Σ) parametrization with locked gate (Pre-Registered):

$$g(a) = \frac{1}{2}[1 + \tanh((a - a_t)/\delta_a)] \quad (1)$$

$$f_k(k) = [1 + (k/k_c)^2]^{-1} \quad (2)$$

$$\mu(k, a) = 1 + A_\mu \cdot g(a) \cdot f_k(k) \quad (3)$$

$$\Sigma(k, a) = 1 + A_\Sigma \cdot g(a) \cdot f_k(k) \quad (4)$$

Locked parameters (NOT free in MCMC):

Parameter	Value	Source
a_t	0.7	P1 pre-registration (figshare.32037990)
δ_a	0.08	EFC v3.11 ledger, regime-transition width
k_c	0.05 h/Mpc	P3 pre-registration (figshare.32023788)

Free EFC parameters:

Parameter	Prior	Range	Motivation
A_μ	Uniform	$[-0.20, 0.05]$	EFC predicts $A_\mu < 0$; small positive for null-test symmetry
A_Σ	Uniform	$[-0.05, 0.20]$	EFC predicts $A_\Sigma > 0$; small negative for null-test symmetry

EFC-VariantH slip constraint (optional secondary test): Enforce $\Sigma(k, a) = 1 - \alpha \cdot (\mu(k, a) - 1)$ with $\alpha = 1.0$ locked. This reduces to a one-parameter EFC extension of Λ CDM.

1.3 Pixel-Level Check

Any implementer must verify before running full analysis:

- When $A_\mu = A_\Sigma = 0$: EFC reduces to Λ CDM identically. Forward model outputs ($\xi_\pm$, $C_\ell^{\kappa\kappa}$, $f(z)$, σ_8) must match Λ CDM within numerical precision ($\leq 0.01\%$).
- When $a_t \gg 1$ or δ_a very small with $A_\mu, A_\Sigma \neq 0$: $g(a) \rightarrow 0$ for all observable z ; EFC $\rightarrow \Lambda$ CDM. Same verification.

If pixel-level check fails: STOP. Pipeline has implementation error.

1.4 Phenomenological-to-action mapping disclosure (added v1.1)

The (A_μ, A_Σ) parametrization used in §1.2 is the MGCAMB/MGCLASS-native phenomenological form, chosen for likelihood-pipeline compatibility. This parametrization is NOT directly derived from the EFC action (DOI 10.6084/m9.figshare.31876324). The action-level parameters $K_0, V', \alpha, \rho_{\text{crit}}, \gamma_0$ of the sealed lensing engine (`lensing.py`, DOI 10.6084/m9.figshare.32037990) are not in one-to-one correspondence with (A_μ, A_Σ) .

Scan analyses on the sealed lensing engine (companion compound paper, §3.3–3.4) indicate that:

- **K_0 is the primary control parameter** for the lensing-sector crossover redshift: $z_{\text{crossover}}$ varies by $\Delta z \approx 0.87$ across $K_0 \in [0.5, 4.0]$.
- **Amplitude rescaling** (applied jointly to α, V', γ_0) produces only $\Delta z \approx 0.11$ across a $\times 50$ range.
- The phenomenological-to-action mapping $(A_\mu, A_\Sigma) \leftrightarrow (K_0, V', \alpha)$ is **not yet constructed in the reference EFC implementation**.

A positive §3.1 fast-scan result (“EFC-like perturbations improve DES Y6 fit around Λ CDM best-fit”) must therefore be interpreted as evidence that the phenomenological quadrant consistent with EFC predictions is preferred by data — **not as direct validation of the EFC action**. Correspondingly, a null §3.1 result falsifies the phenomenological (A_μ, A_Σ) direction tested, but does not uniquely falsify the EFC action.

This disclosure matches the open-task acknowledgment in the companion compound paper (P1×P2 cross-observable consistency, DOI 10.6084/m9.figshare.32080059).

2 Data Specification

2.1 DES Y6 Cosmic Shear

Primary test dataset.

- **Source:** DES Y6 Cosmology Results, Abbott et al. 2026
- **Data product:** Cosmic shear two-point correlation function $\xi_{\pm}(\theta)$
- **Format:** SACC file format (`des_y6_shear.sacc`)
- **Tomographic bins:** 4 (Y6 standard)
- **θ range:** per official DES Y6 scale cuts (do not modify)
- **Covariance:** official Y6 analytic covariance including CMB lensing cross-correlation contributions

Critical: use *exactly* the DES Y6 fiducial scale cuts. No expansion to smaller scales. No modification of angular binning.

2.2 KiDS Legacy

Cross-pipeline validation dataset.

- **Source:** Wright et al. 2025, KiDS Legacy cosmological constraints
- **Data product:** Band powers C_{ℓ}^{ij} OR $\xi_{\pm}(\theta)$, whichever is released as primary
- **Tomographic bins:** 6 (KiDS Legacy standard)
- **Scale cuts:** KiDS Legacy fiducial (SOM-based, different from DES)

Critical: do NOT attempt to unify scale cuts between DES Y6 and KiDS Legacy. The cross-pipeline test in §3.3 specifically tests *whether EFC is stable across independent pipeline choices*.

2.3 Nuisance Parameters

Applied identically to both models (Λ CDM and EFC).

DES Y6:

Parameter	Prior	Notes
A_{IA} (NLA amplitude)	Uniform $[-5, 5]$	NLA is DES Y6 fiducial
η_{IA} (NLA z -evolution)	Uniform $[-5, 5]$	
A_{baryon} (HMCode)	Uniform $[2.0, 3.13]$	HMCode 2020 parametrization
m_i (shear cal, 4 bins)	Gaussian $\pm\sigma_{m,i}$	Y6 official values
Δz_i (photo- z shift, 4 bins)	Gaussian $\pm\sigma_{z,i}$	SOMPZ+WZ-derived

KiDS Legacy:

Nuisance handling: numerically profiled in §3.1 fast-scan (fixed nuisance per grid point set to best-fit Λ CDM values), fully marginalized in §3.2 and §3.3 MCMC.

Parameter	Prior	Notes
$A_{\text{IA}}, \eta_{\text{IA}}$	KiDS fiducial NLA or TATT	Match KiDS publication
A_{baryon}	KiDS fiducial prior	
m_i (shear cal, 6 bins)	Gaussian $\pm\sigma_{m,i}$	KiDS Legacy values
Δz_i (photo- z shift, 6 bins)	Gaussian $\pm\sigma_{z,i}$	SOM-based

3 Test Protocol

3.1 Fast-Scan Sensitivity Probe

Purpose: Determine if DES Y6 $\xi_{\pm}(\theta)$ responds measurably to EFC-perturbations around Λ CDM best-fit.

Procedure:

1. Fix all Λ CDM cosmological parameters to Planck 2018 TT+TE+EE+lowE+lensing best-fit values: $\omega_b = 0.02237$, $\omega_c = 0.1200$, $100\theta_{\text{MC}} = 1.04092$, $\tau_{\text{reio}} = 0.0544$, $\ln(10^{10}A_s) = 3.044$, $n_s = 0.9649$.
2. Fix nuisance parameters at DES Y6 best-fit values from Λ CDM-only fit.
3. Scan EFC free parameters on grid: $(A_{\mu}, A_{\Sigma}) \in [-0.12, 0.0] \times [0.0, 0.12]$, $25 \times 25 = 625$ points. At each point: compute χ^2 against DES Y6 $\xi_{\pm}(\theta)$ with fixed covariance.
4. Also scan VariantH 1D line: $A_{\mu} \in [-0.12, 0.0]$, 25 points, with $\Sigma = 1 - (\mu - 1)$ enforced.

Required outputs: χ^2 surface as 25×25 array, saved as `.fits` or `.npz`; χ^2 at origin ($A_{\mu} = A_{\Sigma} = 0$) as reference; $\Delta\chi_{\text{grid}}^2 = \chi_{\text{grid}}^2 - \chi_{\text{origin}}^2$; best-fit grid point; gradient vector at origin; VariantH $\chi^2(A_{\mu})$ as 25-point array; diagnostic 2D heatmap.

Stopping rule: If $|\Delta\chi_{\text{best-fit}}^2| < 1$ and $\|\nabla\chi^2\| < \sigma_{\text{numerical}}$, EFC perturbation is below DES Y6 sensitivity at background-fixed level. Full MCMC (§3.2) is NOT JUSTIFIED. Publish §3.1 result as null-sensitivity finding. §4 falsifier Rule 1 activates.

3.2 Full Model-Comparison MCMC

Purpose: Symmetric EFC vs Λ CDM comparison with all parameters free, only invoked if §3.1 shows measurable sensitivity.

Procedure:

1. Run Λ CDM MCMC against DES Y6: vary 6 cosmological + all nuisance. Sampler: Cobaya MCMC with Gelman-Rubin $R < 0.01$. Minimum chain length: 200,000 samples post-burn-in.
2. Run EFC MCMC against same DES Y6 likelihood: vary 6 cosmological + $\{A_{\mu}, A_{\Sigma}\}$ + same nuisance. Same sampler settings. Initialization: start from §3.1 best-fit grid point with small scatter.
3. Repeat 1 and 2 for EFC-VariantH (1 extra parameter).

Required outputs: $\chi^2_{\min}$ for Λ CDM, EFC-2param, EFC-VariantH; ΔAIC , ΔBIC with explicit parameter counts; full posterior chains (archived as `.txt` or `HDF5`); posterior plots for key parameters $\{S_8, \Omega_m, A_\mu, A_\Sigma\}$; consistency check vs §3.1 gradient direction.

Stopping rule: If $|\Delta\text{AIC}(\text{EFC} - \Lambda\text{CDM})| < 2$, DES Y6 alone does not distinguish models. Report as “statistically indistinguishable”. §3.3 still runs for robustness.

3.3 Cross-Pipeline Test (KiDS Legacy)

Purpose: Determine whether DES Y6 $\leftrightarrow$ KiDS Legacy S_8 divergence is pipeline-signal or model-signal.

Procedure: Repeat §3.2 MCMC against KiDS Legacy (both models) with KiDS nuisance priors. Compute pipeline-shift metrics $\Delta\chi^2_{\text{pipeline}}(\Lambda\text{CDM})$ and $\Delta\chi^2_{\text{pipeline}}(\text{EFC})$, and S_8 shift between pipelines in each model.

Interpretation (binding):

- $\Delta\chi^2_{\text{pipeline}}(\Lambda\text{CDM}) \approx \Delta\chi^2_{\text{pipeline}}(\text{EFC})$ within 1σ : pipeline divergence is *model-independent*. Neither model gains robustness advantage.
- $\Delta\chi^2_{\text{pipeline}}(\text{EFC})$ smaller at $> 2\sigma$: EFC shows robustness advantage; reportable.
- $\Delta\chi^2_{\text{pipeline}}(\text{EFC})$ larger: EFC is *more* pipeline-sensitive. §4 Rule 3 activates.

4 Symmetric Falsification Rules (Binding)

Rule 1 — Sensitivity null (applied after §3.1)

If §3.1 returns flat χ^2 surface ($|\Delta\chi_{\text{best-fit}}^2| < 1$):

- The phenomenological (A_μ, A_Σ) direction with locked gate parameters ($a_t = 0.7$, $\delta_a = 0.08$, $k_c = 0.05$ h/Mpc) is NOT detectable in DES Y6 at background-fixed level.
- §3.2 MCMC is NOT run.
- Companion document’s §§6.2-6.4 claims are withdrawn pending alternative test paths.
- Publication of §3.1 null result is mandatory.
- **Scope of null result (v1.1):** The null falsifies the specific phenomenological direction tested, not the underlying EFC action. Alternative (A_μ, A_Σ) directions or gate configurations remain untested until the phenomenological-to-action mapping is constructed (see §1.4 and companion compound paper).

Rule 2 — Model rejection (applied after §3.2)

If §3.2 returns $\Delta\text{AIC}(\text{EFC} - \Lambda\text{CDM}) \geq +5$ with:

- Chain convergence $R < 0.01$ ✓
- Prior sensitivity test passed (changing A_μ, A_Σ prior ranges by $2\times$ gives same posterior)
- Nuisance marginalization complete ✓
- No pixel-level check failure in §1.3 ✓

Then EFC’s cosmological claims are WITHDRAWN from ledger and public documentation. The RCMP methodological observation (companion document §§2–4) remains valid as epistemic-holism observation but shall not be used to defend EFC cosmology positions.

Rule 3 — Robustness failure (applied after §3.3)

If §3.3 returns $\Delta\chi_{\text{pipeline}}^2(\text{EFC}) > \Delta\chi_{\text{pipeline}}^2(\Lambda\text{CDM})$ at $> 2\sigma$:

- EFC’s implicit claim to pipeline-robustness is WITHDRAWN.
- Future EFC papers must explicitly state “EFC shows higher pipeline sensitivity than ΛCDM on cosmic shear” in abstract.

Publication commitment

All three stages publish regardless of outcome. Cherry-picking favorable results while suppressing null or unfavorable results constitutes scientific misconduct under this pre-registration.

5 Implementation Resources

5.1 Required software

- **MGCAMB** (recommended): github.com/sfu-cosmo/MGCAMB — Hojjati, Pogonian et al.
- **OR MGCLASS II**: github.com/MiguelZuma/MGCLASS_public — Zumalacarregui et al.
- **Cobaya**: cobaya.readthedocs.io — Torrado & Lewis
- **DES Y6 likelihood module**: released with DES Y6 cosmology papers
- **KiDS Legacy likelihood module**: released with Wright et al. 2025
- **GetDist**: for posterior plotting

5.2 Required resources

- CPU time: 5,000–20,000 core-hours for full §3.2 + §3.3 MCMC
- Storage: ~ 10 GB for chain archives
- Reproducibility: container image (Docker/Singularity) recommended

5.3 Estimated timeline for external implementer

With active MGCAMB + Cobaya experience: §3.1 Fast-Scan 2–3 weeks; §3.2 Full MCMC 4–6 weeks; §3.3 Cross-pipeline 4–6 weeks. Total: 3–4 months with clean cluster access.

5.4 Reproducibility artifacts (required)

Required for publication under this specification:

1. Implementation code (MGCAMB patch or MGCLASS config) archived at permanent URL
2. Cobaya YAML config files for all three stages
3. Chain archives (compressed HDF5) at permanent URL
4. Environment specification (Docker/Singularity image hash, OR `conda env export`)
5. Run logs showing pixel-level check results (§1.3)
6. Diagnostic plots listed in §§3.1, 3.2, 3.3

6 Call for Implementation

This specification is pre-registered at Figshare under DOI 10.6084/m9.figshare.32080083 (assigned 2026-04-21) with SHA-256 content hash. Any researcher with access to MG-CAMB/MGCLASS + Cobaya + DES Y6 and KiDS Legacy likelihoods who implements this specification faithfully is invited to:

1. Execute all three stages (§§3.1–3.3)
2. Publish results regardless of outcome per §4 Publication commitment
3. Co-author the resulting paper with the specification author (Morten Magnusson, ORCID 0009-0002-4860-5095), who contributes the pre-registered specification, methodological framework (RCMP), and EFC model development.

Contact: Morten Magnusson, Symbiose Research, Sandnes, Norway. ORCID: 0009-0002-4860-5095. Email: morten@magnusson.as.

What is offered to the implementer:

- A locked, pre-registered, falsifiable test that cannot be revised post-data
- Co-authorship on the resulting paper with clear contribution statement
- Full access to EFC methodology documentation, Ledger, and RCMP framework
- A defensible project: the specification is neutral; whether it supports or refutes EFC is the empirical question

What is required from the implementer:

- Faithful execution of this specification without modification to §§1–4
- Publication of the result regardless of favorability to EFC
- Time and computational resources for the test

7 Version and Lock Information

- **Version:** 1.1
- **Status:** Pre-registered, locked; v1.1 amendments in §1.4 (phenomenological-to-action disclosure), §4 Rule 1 (scope-clause), and §7 (companion papers) do not modify test protocol §§2–3 or falsifier rules, only clarify scope.
- **Lock commitment:** No modifications to §§1.1–1.3 or §§2–3 after publication. Scope-clarifying additions in §1.4 and §4 Rule 1 (v1.1) do not alter the pre-registered test; revisions to core protocol issue new DOI with explicit diff.

Companion papers:

- RCMP methodology: [figshare.31222900](https://figshare.com/figures/datasets/figshare.31222900)
- EFC regime architecture: [figshare.31112536](https://figshare.com/figures/datasets/figshare.31112536)

- EFC action (underlying lensing engine): figshare.31876324
- P1–P6 predictions: figshare.32037990, 32013156, 32023788, 32013738, 31333414
- Tech Note II (μ -based σ_8 suppression): figshare.31333600
- Methodological convergence observation: **figshare.32080080** (published 2026-04-21)
- P1×P2 compound cross-observable consistency paper: **figshare.32080059** (published 2026-04-21)

8 Abbreviated References

- DES Collaboration, Abbott et al. 2026, DES Y6 Cosmology Results
- Wright et al. 2025, KiDS Legacy cosmological constraints, A&A
- Magnusson 2026, The Regime-Consistent Measurement Principle, figshare.31222900
- Magnusson 2026, EFC Action, figshare.31876324
- Magnusson 2026, EFC Predictions P1–P6, figshare.32037990 and related
- Magnusson 2026, Tech Note II, figshare.31333600
- Magnusson 2026, Compound P1×P2, figshare.32080059 (published 2026-04-21)
- Hojjati, Pogosian, Zhao 2011, MGCAMB, JCAP 1108:005
- Zumalacárregui et al. 2016, MGCLASS, JCAP 1708:019
- Torrado & Lewis 2021, Cobaya, JCAP 05:057

End of specification.